Caso Garlasco: digital forensics timeline

A technical-legal analysis of the IT and telecommunications turning points in the two levels of judgment — first instance (GUP Vigevano, 17 December 2009) and appeal (Corte di Assise di Appello di Milano, 6 December 2011) — with the operational digital forensics lessons that emerge from the case.

DIGITAL FORENSICS | PROCEDURAL LAW | CASE STUDY

Scope of the analysis

This presentation reconstructs **exclusively the technical-informatic and telecom** turning points that enter the judicial reasoning, examining how they are handled in terms of admissibility, reliability, probative value, and any evidentiary reopening.

First instance

GUP judgment of the Vigevano court — **17 December 2009** — abbreviated trial with ex officio expert assessment under art. 441 co. 5 c.p.p.

Second instance

Judgment of the Milan Court of Assizes of Appeal — **6 December 2011**, filed **5 March 2012** — full confirmation of the first instance.

Digital alibi

Use of the laptop on the morning of 13/08/2007: pornography sequence → thesis and related timestamps/artifacts.

Telecom forensics

Calls, CDR records, anonymous subscriber, discrepancies in device storage.

Alarm log

Electronic timestamp of the alarm system deactivation: 09:12 on 13/08/2007.

Digital content

Pornographic archives and the discussion of their motivational relevance in the judicial reasoning.

13 august 2007: the digital data of the alibi

09:12 a.m. — The alarm log: the "temporal pivot"

The electronic data

Disarming of the Poggi home alarm system at **09:12 a.m.** on 13 August 2007.

This electronic timestamp is regarded by both rulings as a "hard" datum, meaning it is not disputed in its existence or time.

Why is it decisive?

The 09:12 log serves a threefold function in the judicial reasoning:

- **Sign of life**: attests to someone's presence in the Poggi home at that moment.
- **Temporal window**: defines the compatible margin for placing the homicidal act.
- **Evidentiary hinge**: connects the forensic reconstruction with the digital one (PC and phone calls), creating an integrated multi-source timeline.

Both levels of judgment treat this datum as an indispensable reference point for the entire evidentiary structure.

09:35–10:07 — PC startup and content viewing

The technical-computer expert report (Porta–Occhetti panel) reconstructs this time window with precision, making it the most controversial segment in the entire debate over the digital alibi.

1

09:35

Laptop PC activated. The report identifies the startup moment through system artifacts and operating system logs.

2

09:38–10:07

Viewing erotic-pornographic images. The digital artifacts (thumbnails, MRU, temporary files) document this activity with the relevant timestamps.

3

Thesis opened

Immediately after the pornographic session, the undergraduate thesis file is opened. This behavioral sequence will also be assessed from a psychological perspective.

The ruling evaluates this segment not only as technical proof of an alibi, but also in behavioral terms: the "compatibility" of this sequence with a murder committed moments earlier becomes a subject of legal argument.

10:17–12:20 — Working on the thesis: metadata as evidence

The forensic reconstruction

The Porta–Occhetti panel identifies continuous word-processing activity on the thesis during the **10:17–12:20** segment. The technical issue is that the classic trace — temporary files — was compromised due to alterations suffered by the exhibit.

The methodological breakthrough: metadata

The report uses "outside the OS" information — **document metadata** — to make up for the unreliability of temporary files. This is a significant methodological choice: it goes beyond standard artifacts because the standard ones were degraded by contamination.

The judicial outcome

The first-instance judgment treats this reconstruction as **conclusive evidence** of user interaction during the 10:17–12:20 segment, explicitly rejecting the equation "no temporary files = no activity" as technically and logically incorrect in the specific case.

Technical lesson

When primary artifacts are compromised, document metadata (author, last modified date, number of revisions, editing time) can become the most robust residual evidence.

This is one of the most valuable operational lessons from the Garlasco case for the digital forensics investigator.

13 August 2007 — multi-source correlation: PC + phone calls

One of the most methodologically sophisticated aspects of the judicial reasoning is **multi-source correlation**: the expert report and the judge build a cross-referenced timeline from heterogeneous data.

Pauses in PC activity

The expert report identifies windows of inactivity in the computer session: periods in which no user action is recorded in the operating system artifacts.

Missed calls to Chiara's mobile phone

The CDRs (Call Detail Records) record calls to Chiara Poggi's number that went unanswered at times overlapping with the computer pauses.

Probabilistic reasoning on the "anonymous line"

The expert report applies a statistical analysis over the previous 6 months: anonymous lines are examined for usage patterns, timing, and location in order to attribute some calls to Stasi's landline through probabilistic reasoning.

This is a paradigmatic example of how different kinds of data (system logs, CDRs, historical records) can be integrated into a single temporal reconstruction. Its strength depends on the quality of each source — and the contamination of the PC makes this architecture fragile.

14 August 2007 — The delivery of the PC: context and relevance

The fact

The laptop was acquired by investigators on 14 August 2007, the day after the murder. The manner of delivery became the subject of legal debate on multiple levels.

The prosecution's interpretation (rejected on appeal)

One reconstruction claimed that Stasi had voluntarily handed over the PC in order to build a prepackaged digital alibi — an argument supporting the premeditation of the defense strategy.

The Court of Appeal's response

The second-instance ruling **downsized this conjectural reading**, clarifying that the PC was acquired during the **"delegated home search"** of 14 August — it was therefore not an isolated voluntary handover, but part of an institutional and documented police activity.

This contextual clarification is fundamental: it changes the interpretive frame of the handover and the motives attributed to the defendant.

14–29 August 2007 — The "repeated and improper" accesses

This is the **most serious forensic violation** documented in the Garlasco case. The first-instance judgment uses the exact words "repeatedly and improperly" to describe the Carabinieri's accesses to the PC.

Multiple undocumented accesses

The report of 29 August records operations, but later technical checks show activity more extensive than what was documented: the actual number of accesses exceeds the one declared.

Undocumented USB peripherals

USB peripherals are used on the evidence without adequate documentation. Introducing external devices is one of the fastest ways to produce inadvertent writes and metadata alterations.

Access to an external drive

Access is carried out to an external drive connected to the PC, also without proper documentation and without write-blocking measures.

Multiple accesses to the thesis file

The thesis file — a central element of the digital alibi — is opened several times during the pre-forensic phase, potentially altering the last access timestamps.

29 August 2007 — The moment of discontinuity

Before 29 August

"Non-forensic" phase: the exhibit is in the custody of the Carabinieri, subject to repeated access, without write-blocking protections, with incomplete and inaccurate documentation.

Duration: 15 days of exposure to uncontrolled manipulation.

From 29 August onward

Only on 29 August 2007 was the digital exhibit transferred to **specialized consultants** who carried out the **forensic copies** using appropriate methods.

The first-instance judgment defines this transition as the "moment of discontinuity": the clear boundary between non-forensic activity and forensic/consultancy operations.

The damage, however, had already occurred. The Porta–Occhetti expert report had to work on an exhibit that was already degraded, trying to reconstruct the truth "despite" contamination — not thanks to a proper chain of custody.

The order of 17 March 2009: rejection of inadmissibility

The defense raises the inadmissibility of the PC contents, relying on the documented contamination. The GUP rejects the objection with the order of 17 March 2009, developing an articulated procedural theory of digital evidence.

1

Intrinsic fragility of the digital document

The court explicitly refers to the nature of the digital document as an element that can be easily altered and deleted, distinguishing it from traditional paper evidence.

2

Need for safeguards and reference to L. 48/2008

The judge connects the necessary safeguards (faithful and non-modifiable copies) to the legislative evolution of L. 48/2008, which reformed the means of evidence collection with an emphasis on preservation and the prevention of alterations.

3

Qualification of the activity: PG, not technical examination

Key point: the court classifies the Carabinieri's activity as **"PG acts" under arts. 55 and 348 c.p.p.** (collection of useful elements), and not as a technical examination under arts. 359/360 c.p.p. — for which a specific question, specialized expertise, and a final report would have been required.

The classification as "good-faith PG acts" is the mechanism that allows the judge to admit the material while still recognizing its methodological compromise. Not inadmissible, but to be assessed with the reservations imposed by contamination.

The ex officio expert examination: response to contamination

The judicial logic

The judge directly links the documented contamination to the need for a technical expert examination: the loss and alteration of traces can harm both the prosecution and the defense, compromising "the neutral value of the ascertainment of truth".

From this premise comes the decision to order a **technical-computer expert examination under art. 441 co. 5 c.p.p.** in abbreviated trial proceedings — a typically residual tool, here used to "repair" the epistemic damage caused by the pre-forensic phase.

The Porta–Occhetti panel

The expert panel's task is to reconstruct the truth despite contamination: a work of digital archaeology on degraded data, using every residual artifact, metadata, and alternative source available.

The two key outcomes

1. Quantification of the damage: "73.8% of visible files" altered or removed (over 56,000 files). Effects described as "devastating" for overall integrity and for reconstructing the timeline of events.

2. Residual reconstruction: Despite the damage, the login → pornography → thesis sequence is reconstructed through residual artifacts and metadata, closing the "evidentiary gap" left by the unreliability of temporary files.

Stasi's cellphone: late acquisition and authenticity

The problem: 22 months of latency

Alberto Stasi's cellphone — relevant for alarms and morning routines — was acquired in the expert report only on **18 June 2009**, almost 22 months after the murder. The judgment stresses that the failure to inspect it immediately creates a structural problem of authenticity.

Available data and plausibility assessments

Even without certainty, the judgment evaluates circumstantial elements: last recorded use on 22 August 2007; battery drained upon delivery; alarm functions disabled; automatic power-on function active. The judge makes assessments of **plausibility**, not technical certainty.

The lesson: mobile devices are dynamic

A mobile device not inspected immediately may change its contents in unpredictable ways: updates, synchronizations, automatic deletions, battery drain with loss of volatile data. NIST SP 800-101r1 is explicit about the need for immediate preservation and validation through hashes.

Appeal: confirmation and "no" to renewed forensic review

On appeal, the relevant grounds of complaint in the field of computer evidence are discussed by the Court on three distinct levels.

1

Criticism of the behavioral assessment

The defense disputes the way the first judge assessed the pornography, the reticence, and the "psychological compatibility" with the homicide.

2

Criticism of the digital alibi and the handover of the PC

The logical soundness of the forensic reconstruction and the correctness of the reasoning regarding the delivery of the computer in the context of the search are challenged.

3

Request by the PG for technical follow-up

The Prosecutor General requests further analysis of the PC "with more up-to-date techniques" — but this request is then abandoned in the closing submissions.

Outcome: The Court does not order any renewed digital evidentiary examination, noting that the request "with more up-to-date techniques" lacks specific references and is not pursued in the closing submissions — so "there is no need to deal with it." The judgment of 17 December 2009 is fully **upheld**.

CHAPTER 4

The 10 errors of digital forensics: impact map

The Garlasco case is a remarkable case study for digital forensics training. Below, the 10 most obvious errors emerging from the file are analyzed, ordered by forensic impact and linked to the principles of the main international guidelines (ACPO, SWGDE, NIST).

Critical errors (1–3): the fundamental violations

🔴 Error 1: Working on the original without protections

“Repeated and improper” access to the PC between 14 and 29 August 2007, with effects measured in **73.8% of visible files** altered or destroyed. Direct denial of **ACPO Principle 1**: “no action should change data relied upon in court”.

🔴 Error 2: Failure to carry out an immediate forensic duplicate

The forensic duplicate (bitstream image) arrived only on 29 August, after 15 days of manipulation. Direct violation of **NIST SP 800-86**: if the material may be needed for proceedings, it must be acquired immediately on a copy, the original must be preserved, and every step documented.

🔴 Error 3: Lack of write-blocking

No hardware or software write-blocker was used during access to the item. **SWGDE** is explicit: minimizing changes to source data is typically achieved with write-blockers. The judicial reconstruction documents file creation and alteration during custody.

Serious errors (4–6): documentation, expertise, peripherals

🟠 Error 4: Incomplete documentation (weak audit trail)

The report of 29 August does not match what was later technically established: the actual number of accesses was higher, USB peripherals were not indicated, access to an external disk was omitted, and multiple accesses to the thesis file were not recorded. Violation of **ACPO Principle 3** (repeatable audit trail) and SWGDE best practices on chain of custody.

🟠 Error 5: Untrained operators without technical assistants

The judgment notes that the operations were carried out by law enforcement officers who were "not experts in the field" and without the requirements of a technical examination (question, specific expertise, report). **ACPO Principle 2** requires competence and the ability to explain actions and implications when accessing the original.

🟠 Error 6: Contamination from USB peripherals and external drives

Use of USB peripherals and access to external drives without tracing or protection. Connecting external devices can introduce automatic writes (autorun, mount, drivers, system metadata). **SWGDE** requires understanding and documenting the impact of every chosen technique, making processes auditable and repeatable.

Medium-high errors (7–10): governance and method

1 Late acquisition of the cellphone (22 months)

The phone acquired in June 2009 raises the issue of the authenticity of the content. **NIST SP 800-101r1** on mobile forensics emphasizes the need for immediate preservation and hash validation, since devices are dynamic and change over time.

2 Fragile technical inferences (poorly selected artifacts)

The expert critique of the equation "no temporary files = no activity" shows how contamination of the evidence produces a cascading effect: degraded data generate fragile inferences, which in turn make the entire analytical reconstruction unstable.

3 Unstructured digital investigation governance

Lack of a clear Plan → Capture → Analyse → Present pipeline and of a defined case officer responsibility (**ACPO Principle 4**). The first judge describes "radical confusion" in the management, with "metastatic" effects on the examination.

4 Lack of timeline standardization (time skew, time zone)

The case goes so far as to consider the hypothesis that the times on the PC did not correspond to real time (time skew, clock drift). The expert report must explicitly rule this out. **NIST SP 800-86** requires explicit verification of temporal integrity as part of the robustness of the forensic reconstruction.

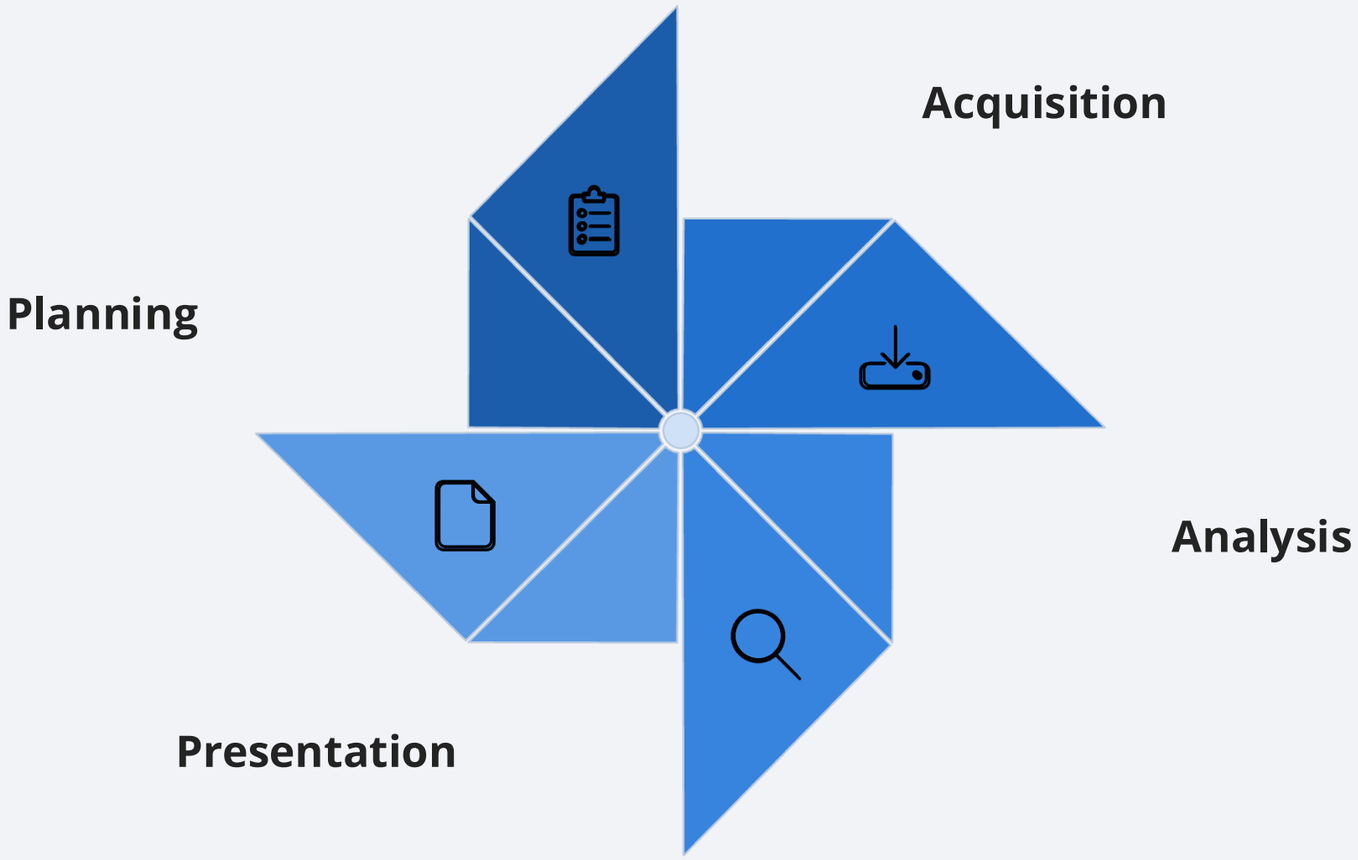
Error matrix → international standards

A mapping between the errors in the Garlasco case and the principles of the reference guidelines in international digital forensics.

Error / violation	ACPO	SWGDE	NIST
Working on the original without protections	Principle 1	§ Minimization	SP 800-86
Immediate bitstream image not made	Principle 1	§ Acquisition	SP 800-86
Absence of write blocker	Principle 1	§ Write-blocking	SP 800-86
Incomplete audit trail / inaccurate report	Principle 3	§ Documentation	SP 800-86
Unqualified operators	Principle 2	§ Competence	—
USB / peripherals not tracked	Principle 3	§ Auditability	SP 800-86
Late cellphone acquisition	Principle 1	§ Integrity	SP 800-101r1
Inferences from degraded artifacts	Principle 4	§ Analysis	SP 800-86
Missing investigative governance	Principle 4	§ Pipeline	SP 800-86
Unverified time skew	—	§ Repeatability	SP 800-86 § 3

The correct model: the standard forensic pipeline

What should have happened on August 14, 2007? The comparison with international best practices outlines a precise and non-negotiable forensic pipeline.



Every phase has a precise documentary output: in the Garlasco case, the "Capture" phase is essentially absent for the first 15 days, making all subsequent phases intrinsically vulnerable to challenges regarding the integrity of the data.

Digital evidence in the two degrees: synoptic comparison

First degree (GUP Vigevano, 2009)

Key decision: Rejection of inadmissibility (order of 17/03/2009).
Qualification of the judicial police: Acts under arts. 55 and 348 c.p.p., not technical assessment.
Response to contamination: Court-appointed expert examination (Porta–Occhetti).
Damage assessment: 73.8% of visible files altered.
Reconstruction: PC sequence reconstructed “despite” contamination, via metadata.

Second degree (CAA Milano, 2011)

Key decision: No renewed IT evidentiary hearing.
On the alibi from the computer: First-degree reasoning confirmed.
On the PC handover: The “spontaneous/strategic” reading was scaled back — it was a delegated search.
On behavioral critiques: Conjectural readings about pornography rejected.
Outcome: Full confirmation of the judgment of 17/12/2009.

The "holding" of the Garlasco case for digital forensics

> "The Carabinieri's activity was **methodologically incorrect**, but carried out in **good faith**: it does not constitute technical evidence under arts. 359/360 c.p.p., but rather police acts under arts. 55 and 348 c.p.p. Contamination does not make the material inadmissible, but it does require a restorative expert examination and a critical assessment of the evidentiary weight of degraded artifacts."

— Summary of the ratio decidendi of the GUP of Vigevano (ordinance 17 March 2009 + judgment 17 December 2009)

For the judge

Contamination does not automatically equal inadmissibility. However, it does impose a heightened duty to provide reasons and, where possible, a restorative technical examination.

For the defense

Challenging the chain of custody is fertile ground, but it does not guarantee exclusion of the evidence: one must show that the contamination altered the data in a way unfavorable to the defendant.

For the prosecutor and the police

"Good faith" protects against procedural sanctions, but it does not restore the integrity of the data. The cost of non-forensic procedures is paid in the credibility of the evidence, not in formal inadmissibility.

Operational lessons: what changes with L. 48/2008

The regulatory context cited by the GUP

The first-instance judgment explicitly refers to **Law 48/2008** (ratification of the Budapest Cybercrime Convention) as a regulatory evolution that strengthens safeguards in the acquisition of digital evidence.

The changes introduced by L. 48/2008 to the means of searching for evidence (arts. 244, 247, 254, 259, 260, 352, 354 c.p.p.) introduce:

- Obligation to create **forensically sound, non-modifiable copies** of data.
- Obligation to adopt measures suitable to ensure the **preservation of original data**.
- Obligation to prevent **alterations** during acquisition operations.
- Obligation to document the operations performed.

The temporal paradox

Accesses to the PC took place in **August 2007**. L. 48/2008 entered into force in **March 2008**.

The GUP uses the law not as a standard for the legality of prior acts, but as a **regulatory confirmation** of a principle already inherent in the system: the need for safeguards in the acquisition of digital evidence was a methodological duty before it became an explicit legal obligation.

This distinction is essential to understanding the rationale behind the rejection of inadmissibility: there was not yet a norm violated in a technical sense, but rather a best practice that was ignored.

Takeaways for the forensic professional

Isolate first, analyze later

The original evidence must not be touched. Ever. The bitstream image is the starting point of every analysis — not an option.

Document every action

The audit trail must be more detailed than you think is necessary: number of accesses, times, tools used, connected peripherals, operators present. Every discrepancy will be exploited in court.

Hash always, immediately, and again

Calculate the hash of the original before and after acquisition. Calculate the hash of the copy. Document everything. Integrity is proven, not claimed.

Volatility does not wait

Mobile devices, RAM, system logs: every hour that passes without acquisition means data lost or altered. Delay costs authenticity.

Conclusions: the Garlasco case as a digital forensics laboratory

The Garlasco case remains, more than fifteen years after the events, the most complex Italian case study of **digital evidence in court**. The lessons it offers are still fully relevant today.

Contamination has a measurable cost

73.8% of visible files altered: the forensic damage can be quantified, and this quantification itself becomes evidence in court.

Contamination does not mean inadmissibility

The Italian legal system does not automatically exclude contaminated digital evidence: it reduces its weight, requires careful evaluation, and may call for remedial expert analysis.

Metadata as a last resort

When standard artifacts are compromised, document metadata and alternative sources become the most robust remaining evidence.

International standards are the benchmark

ACPO, SWGDE, NIST: these guidelines have no force of law in Italy, but judges implicitly use them as the benchmark for "proper" forensic practice.

Primary sources: Sentenza GUP Tribunale di Vigevano, 17 December 2009 · Sentenza CAA Milano, 6 December 2011 (published 5 March 2012) · ACPO Good Practice Guide for Digital Evidence (2012) · SWGDE Best Practices · NIST SP 800-86 · NIST SP 800-101r1